

Schur Functions at $(1, -1, t, t^{-1})$

*A closed form on the non-identity component of $O(4)$:
three $SU(2)$ characters read off the 2-quotient*

Part IV — a vanishing criterion with no hypothesis on the shape of the partition, the row that is not constant, and an open problem its owners declared

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July 26, 2026

Abstract

Part III left one item open and said so: whether its classification of the representations whose character vanishes identically on the non-identity component of $O(4)$ was a corollary of some general criterion. It is — and of something stronger than a criterion. For every partition λ with at most four rows, $s_\lambda(1, -1, t, t^{-1})$ is $\pm$ a product of *exactly three* $SU(2)$ characters, read off the 2-quotient of λ , or zero, with no hypothesis whatever on the shape of λ ; the vanishing classification is precisely its zero locus. The consequence is a compression: conditionally on this identity, an arbitrarily large four-row partition collapses to three integers with no loss of anything the physics of Part III can see, so that every quantity surviving its leading cancellation factors through the 2-quotient and weight enumeration is replaced by explicit polynomial formulae. The alphabet $\{1, -1, t, t^{-1}\}$ is the generic semisimple element of $O(4)^-$, and it sits between two factorisation programmes and belongs to neither: the reciprocal-pair line of Ciucu–Krattenthaler and Ayer–Behrend, whose engine needs every inversion-fixed point of the alphabet to contribute a *constant* row of the bialternant — which $+1$ does and -1 does not — and the root-of-unity line of Littlewood, Prasad, Ayer–Kumari and Karmakar, whose alphabets carry a free variable in every letter. Replacing the two fixed-point rows by even/odd column indicators repairs the argument at rank one and does not at rank two, which is where we stop and say so. We state the identity as *verified* and not as a theorem: it is checked exhaustively by two independent implementations on all 3060 partitions with four parts ≤ 14 and all 4845 with parts ≤ 16 , with no mismatch and no residue in the zero locus, and no proof is given. Ciucu–Krattenthaler wrote in 2008 that they could not offer a representation-theoretic interpretation of their factorisations, and Ayer–Behrend and Ayer–Kumari repeated it in 2018 and 2025; the reading offered here — a character on the non-identity component — is of that kind, conditional on the identity being true, and it reaches the factorisation and not only the vanishing. *Added in version 3*: that reading has a home in the literature we failed to cite in version 1. Characters of a disconnected group on a non-identity component are *twining characters*, and the identity that computes them by folding is due to Jantzen [17] and, in the form closest to ours, to Kumar–Lusztig–Prasad [18]; our restriction is reducible, so it is reached from theirs by one step of Clifford theory. The mechanism is theirs and we credit it here; what this paper contributes is the explicit evaluation at free t . What we take to be the result of this paper, however, is neither the identity nor that reading, but what they make possible: $\Phi = F \circ Q$, the physical functional factoring through the 2-quotient. The Schur identity is the door; the reduction is what is behind it.

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1 The item we left open

Part III classified, in $SU(4)$ Dynkin labels, the representations for which

$$D(t) = s_\lambda(1, -1, t, t^{-1}) \quad (1)$$

vanishes identically. The classification was exact, it was certified in Lean, and it was honest about what it owed: the *mechanism* is classical — an $O(2n)$ irreducible representation and its associate $\mu \otimes \det$ have opposite characters on the non-identity component (the *improper* component, in the older terminology), so self-associate ones contribute nothing there — and what was claimed was the arithmetic, not the phenomenon. That paper ends by stating four questions it cannot answer. We quote them verbatim from its version 5 [1], resolving in brackets the two cross-references that are internal to it:

Q1 Is $\mathcal{Z}$ a corollary of some general criterion for a $GL(n)$ character to vanish on the non-identity component, or is it particular to $SU(4)$? **Q2** The theorem says where D is zero. *What is D when it is not?* We use its coefficients throughout and never learn what function they belong to. **Q3** Why exactly two branches? [Part III, Figure 6] shows no third cause of vanishing, and we have no reason why there could not be one. **Q4** The moments $\sum_m m^{2r} \delta(m)$ decide everything that survives [Part III, §6], and we obtain them by enumerating the weight system of each irreducible. Is there structure there, or is enumeration the only route?

The four have one answer between them, and it is the subject of this paper:

question	answer, and where
Q1 general criterion?	yes, and by something stronger — §3
Q2 what is D when non-zero?	$\pm$ three $SU(2)$ characters off the 2-quotient — (6)
Q3 why exactly two branches?	a product of three characters has exactly two ways to vanish — §4
Q4 structure in the moments?	δ is a threefold convolution; every moment is polynomial — §5

This paper answers it. The answer is not a criterion. It is a closed form for the character itself, of which the criterion is the locus where the form evaluates to zero.

One word about priority among the four, because it shapes everything after §4. **Q1–Q3** are about the zeros, and the zeros were already classified: Part III proves that outright. **Q4** is about what is left once the leading term has cancelled, and that was the question Part III actually needed answered and could not. So the centre of this paper is not the vanishing criterion but (11) in §5: the physical functional factors through the 2-quotient. The closed form is how that becomes possible; it is the mechanism, not the point (Figure 1).

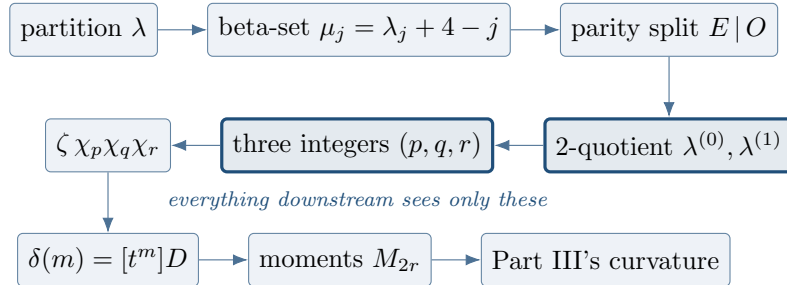


Figure 1: The chain, and where it narrows. Reading left to right and then back, a partition with four rows — of unbounded size — is compressed at the third step into three integers, and nothing downstream ever sees anything else. The two boxed steps are what this paper supplies; the rest is standard. (11) states the narrowing as an identity of functionals.

These two papers should be read as a pair, and the pairing is not merely editorial. Part III is a physics argument that ran into a question about symmetric functions and exported it; this

one answers that question and hands back a tool the physics needed and did not have. The exchange closes in §5: the same factorisation that decides where the character vanishes also makes every moment of its coefficient sequence a polynomial in three integers, and those moments are what Part III was obtaining by enumerating weight systems. A reader interested only in the combinatorics may stop before that section; a reader who came from Part III should start there.

The object is not exotic. Since $1 \cdot (-1) \cdot t \cdot t^{-1} = -1$ and the multiset is stable under $x \mapsto x^{-1}$, the alphabet $\{1, -1, t, t^{-1}\}$ is the eigenvalue multiset of an element of $O(4)$ with $\det = -1$: the generic semisimple element of the non-identity component. So (1) is a $GL(4)$ character restricted to the non-identity component, and the question “when does it vanish, and what is it when it does not” is a question about characters on a disconnected group. That framing is what made the search tractable, and it is also what explains why the answer was not already on the shelf.

2 Two programmes, and the gap between them

Specialisations of Schur functions that factorise are a subject with two active lines, and our alphabet falls between them.

The reciprocal-pair line. Ciucu–Krattenthaler [2] and then Ayer–Behrend [3] factorise s_λ at

$$(X, \bar{X}), \quad (X, \bar{X}, 1), \quad (X, \bar{X}', 1), \quad (2)$$

into products of orthogonal and symplectic characters, for λ self-complementary in a rectangle. Their engine is a single block identity, Ayer–Behrend Eq. (23):

$$\det \begin{pmatrix} A & B \\ B & A \end{pmatrix} = \det \begin{pmatrix} A - B & B \\ B - A & A \end{pmatrix} = \det \begin{pmatrix} A - B & B \\ 0 & A + B \end{pmatrix} = \det(A - B) \det(A + B), \quad (3)$$

reached by scaling rows to remove the rectangle width and reversing the columns of the right block so that the two exponent multisets coincide. The self-complementary hypothesis is consumed *only* in that column reversal, and it says exactly one thing: the beta-set is symmetric about its midpoint.

The root-of-unity line. Littlewood [21], Prasad [8], Ayer–Kumari [4], Kumari [5], Albion [6] and Karmakar [7] work at $(X, \omega X, \dots, \omega^{t-1} X)$ and its variants, where the vanishing is governed by the t -core of λ and the factorisation by the t -quotient.

Ours is in neither. $\{1, -1\} \cup \{t, t^{-1}\}$ is not of the form $(X, \bar{X})$: solving $\{t, t^{-1}\} = \{y, -y\}$ forces $t^2 = -1$, so t is not free. And it is not $\omega^k X$: the letters ± 1 carry no free variable. The natural place for it in (2) is the next member of the sequence,

$$(X, \bar{X}, 1, -1), \quad (4)$$

the case in which *both* fixed points of $x \mapsto x^{-1}$ are present rather than one. That member is missing (Figure 2), and the reason is sharper than an oversight, as §6 shows.

Open flank. Why are these two programmes separate at all? Both compute specialisations at elements of a classical group; one takes the element to have distinct reciprocal eigenvalue pairs, the other to have eigenvalues on a μ_t -orbit. A general alphabet is a multiset of roots of unity times monomials in free moduli, and the two programmes are the two extreme cases of that. *Is there a classification of which such specialisations factorise?* We do not know of one, and the obvious conjecture — that factorisation is controlled by how the element’s centraliser sits inside the group — is easy to state and, on the evidence of §7, not obviously true.

alphabet	det	inversion-fixed points	who did it
$(X, \bar{X})$	+1	none	Ciucu--Krattenthaler 2008
$(X, \bar{X}, 1)$	+1	+1: constant row	Ayyer--Behrend 2018
$(X, \bar{X}, 1, -1)$	-1	-1: parity row	<i>this paper</i>

the block identity needs every fixed point to give a CONSTANT row;
-1 gives $((-1)^{\mu_j})_j$, and the triangularisation stops

Figure 2: The sequence, and the step that was never taken. Each row is an alphabet stable under $x \mapsto x^{-1}$; the third column counts the points that stability leaves fixed, because those are the ones that contribute a row to the bialternant on their own. Adding the first fixed point +1 costs nothing — its row is constant, and the block argument absorbs it. Adding the second changes the determinant of the element from +1 to -1, i.e. moves it off the identity component, and its row is a parity character rather than a constant. Everything in this paper is a consequence of that single row.

3 The closed form

Fix λ with at most four rows and write $\mu_j = \lambda_j + 4 - j$ for its beta-set. Split by parity,

$$E = \{j : \mu_j \text{ even}\}, \quad O = \{j : \mu_j \text{ odd}\},$$

write the even μ_j as $2\alpha_i$ and the odd ones as $2\beta_i + 1$, and let $\lambda^{(0)}, \lambda^{(1)}$ be the partitions with beta-sets α, β — that is, the two components of the 2-quotient of λ [20, Ch. I, Ex. 1.8]. Put

$$\zeta = (-1)^{\text{inv} + |E||O|}, \quad \text{inv} = \#\{i < j : \mu_i \text{ odd}, \mu_j \text{ even}\}, \quad (5)$$

and let $\chi_k(t) = t^k + t^{k-2} + \dots + t^{-k}$, with $\chi_{-1} = 0$.

Observation 1 (verified, not proved). *For every partition λ with at most four rows,*

$$s_\lambda(1, -1, t, t^{-1}) = \begin{cases} 0, & E = \emptyset \text{ or } O = \emptyset, \\ \zeta \operatorname{sgn}(z) \chi_{\lambda_1^{(0)} - \lambda_2^{(0)}} \chi_{\lambda_1^{(1)} - \lambda_2^{(1)}} \chi_{|z|-1}, & |E| = |O| = 2, \quad z = |\lambda^{(0)}| - |\lambda^{(1)}| - 1, \\ \zeta \chi_{\nu_1 - \nu_2} \chi_{\nu_2 - \nu_3} \chi_{\nu_1 - \nu_3 + 1}, & |E| = 3 \text{ or } |O| = 3, \end{cases} \quad (6)$$

where ν denotes the three-part quotient component. In the middle case $z = 0$ gives zero as well, through $\chi_{-1} = 0$.

So the character is always $\pm$ a product of exactly three $SU(2)$ characters, or zero (Figure 3), and — this is the point worth pausing on — **there is no hypothesis on the shape of λ** . Ciucu–Krattenthaler and Ayyer–Behrend must assume self-complementarity; (6) assumes nothing. That difference is not an improvement on their theorem; it is a symptom of a different mechanism, and §6 says which.

Why three, and why $SU(2)$. The alphabet has one free parameter, so whatever the answer is, it is a Laurent polynomial in a single variable, symmetric under $t \mapsto t^{-1}$ — that is, a virtual $SU(2)$ character. The content of (6) is that it is never merely virtual and never irreducible beyond three factors: two of the three come from the two quotient components separately, and the third is a *cross* term measuring how far apart the two components are in size.

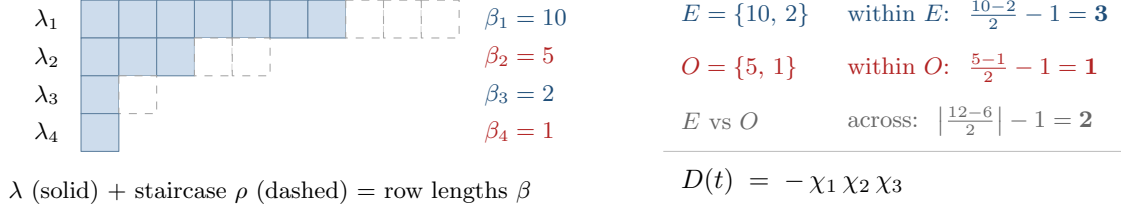


Figure 3: The three spins, read off one diagram. Attaching the staircase $\rho = (3, 2, 1, 0)$ to $\lambda = (7, 3, 1, 1)$ turns the row lengths into the beta-set $\beta = (10, 5, 2, 1)$. Splitting β by parity gives E and O ; the two *within-class* half-differences and the one *cross-class* half-difference, each less one, are the three spins $(3, 1, 2)$, and (5) fixes the sign. No shape hypothesis on λ enters anywhere.

Open flank: why three, and what is the third factor? The first two factors have an obvious owner — one per component of the 2-quotient — and the third does not. It depends only on $|\lambda^{(0)}| - |\lambda^{(1)}|$, and it is the one that produces the second branch of zeros. That dependence is itself a hint about where it lives: an imbalance is not a property of either component but of the *coupling* between them, so if the third factor is attached to anything it is attached to that, and not to $\lambda^{(0)}$ or $\lambda^{(1)}$ separately. Three questions a reader is entitled to ask, none of which we can answer. (i) Is the third factor the character of a multiplicity space? If the restriction to the coset decomposes as $\text{Tr}(\sigma \mid V_{j_L} \otimes V_{j_R} \otimes M)$ with M carrying the imbalance, the identity would stop being combinatorial and become a branching theorem, and $\text{Spin}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$ makes that reading tempting. (ii) Is the count three equal to $2n - 1$ at rank n ? At rank one that reads $3 = 2 \cdot 2 - 1$, two quotient components and one cross term, and it predicts five factors at rank two — which §7 shows is false in general, so if the pattern survives at all it survives on a subfamily. Which one? (iii) Is there a bijective proof? Each χ_k has unit coefficients, so $\delta(m)$ counts lattice points; a bijection between those points and something attached to λ would explain the formula rather than verify it.

Four grades of claim, and which is which. This paper mixes borrowed theorems, an unproved identity, things that follow from it, and things we merely suspect. Conflating them is the easiest way to mislead, so they are graded once, here.

grade	what it means	instances
Proposition	proved, independently of anything here	Prop. 1 (via Part III); both folklore sufficiencies
Observation	exhaustively verified, <i>no proof</i>	(6), the closed form
Inherited	classical, cited, not ours	(9), (12), log-concavity
Open	suspected, with a stated test	the third factor's meaning, rank two, the sign's origin

We state Observation 1 as verified and not as a theorem, and we mean the distinction. It has been checked exhaustively by two independent implementations — one deriving the quotient through Sage's own 2-quotient, one re-deriving α and β from μ directly, so that a convention error in either would show — on all 3060 partitions with four parts ≤ 14 and on all 4845 with four parts ≤ 16 , against Jacobi–Trudi evaluations in exact integer Laurent arithmetic. There were no mismatches. There is no proof.

4 The zero locus is Part III's class

Reading off where (6) evaluates to zero gives two branches, and they are the two clauses of Part III's Theorem: E or O empty is the statement that all four beta-numbers share a parity, and in λ -coordinates that is $\lambda_1 \not\equiv \lambda_2 \not\equiv \lambda_3 \not\equiv \lambda_4$; and $z = 0$ in the balanced branch is $|\lambda^{(0)}| = |\lambda^{(1)}| + 1$, which is λ self-complementary — $\lambda_i + \lambda_{5-i}$ constant — with that constant odd.

Proposition 1 (proved independently in Part III). *The character vanishes exactly on*

$$D \equiv 0 \iff \underbrace{\lambda_1 \not\equiv \lambda_2 \not\equiv \lambda_3 \not\equiv \lambda_4}_{\text{alternating parity}} \quad \text{or} \quad \underbrace{\lambda_i + \lambda_{5-i} = c \quad \forall i, \quad c \text{ odd}}_{\text{self-complementary, odd constant}} \quad (7)$$

This is Part III’s class $\mathcal{Z}$. The two vanishing mechanisms of (6) — an empty parity class, or a balanced 2-quotient whose cross-index z degenerates — are disjoint, even though the two λ -shorthands boxed above are not mutually exclusive as written.

Attribution added in version 3. The first branch is not ours. Setting $t = 1$ sends our alphabet to the order-2 element $(1, -1, 1, 1)$, and our rigidity lemma ($D \equiv 0$ exactly when $D(1) = 0$) carries the criterion there and back. Karmakar [7], Thm. 1A, states precisely that criterion for $\text{GL}(n, \mathbb{C})$ at $C_{n-k,k}$: his $\eta_i(\lambda)$ is our parity split of $\lambda + \rho$, and his normalisation “ $\#\eta_0 \geq \#\eta_1$ after twisting by the determinant character” is our determinant twist. His Thm. 1B goes further and factorises the value at that element through the 2-quotient, constant included — the $t = 1$ shadow of (6). What remains specific to this paper is the second branch and, above all, the evaluation at *free* t . That programme is live: Nadimpalli–Pattanayak–Prasad [19] generalise [7] to torsion elements and pose, as a question of the third author, exactly this shape — vanishing when one centraliser dimension exceeds another, and a constant times a *dimension* when they agree. Our $t = 1$ point is an instance of it. The deformation off that point is not: their programme has no free parameter, and no non-identity component. (We checked, and record the negative: $s_\lambda(1, -1, t, t^{-1})$ is *not* $\pm$ a constant times $s_{\lambda(0)}(1, t, t^{-1})$ — 0 of 105 cases — so the free- t object is not simply the character of the centraliser representation whose dimension Karmakar’s Thm. 1B computes.)

Why this is a proposition and not a corollary. The statement does not depend on Observation 1. Part III proves the classification by a different route and certifies it in Lean, so it stands whatever becomes of (6). What the closed form contributes is not the fact but the *cause*: it exhibits $\mathcal{Z}$ as the locus where the product degenerates, and it does so *conditionally on an identity we have not proved*. Deriving an already-proved theorem from an unproved observation does not promote the observation; it only makes the derivation conditional, and it is worth saying which way the logic runs. A reader who rejects (6) keeps Proposition 1 entire and loses only the explanation.

Open flank: is the dichotomy forced? We can say that there is no third branch; we cannot say why there could not be. In the closed form the two branches are visibly different species — one kills the whole product by emptying a parity class, the other kills a single factor by degenerating its index — and a reader may reasonably ask whether that is a genuine dichotomy or an artefact of rank one. Two probes would settle it. *Is there a duality exchanging the branches?* Both are invariant under conjugation composed with complementation in a suitable rectangle, which is suggestive and which we have not pursued. *And does a third branch appear at rank two?* It does: §7 exhibits a partition that vanishes with both parity classes occupied, which is neither of our two causes. So the dichotomy is a rank-one statement, and calling it a structural fact would be an overclaim.

The decomposition is exact and leaves nothing over. Of the 560 vanishing partitions in the parts ≤ 16 range, 420 fall in the first branch and 140 in the second, and none outside the two (Figure 4).

One sufficiency direction is folklore and we disclaim it. That self-complementarity with odd constant forces vanishing is two lines and is not new: writing $\lambda_i^* = c - \lambda_{5-i}$ one has $s_{\lambda^*}(x) = (x_1 \cdots x_4)^c s_\lambda(x^{-1})$; our alphabet is inversion-stable, so $s_\lambda(p^{-1}) = s_\lambda(p)$; and $x_1 \cdots x_4 = \det = -1$. If $\lambda^* = \lambda$ then $s_\lambda(p) = (-1)^c s_\lambda(p)$, and c odd forces zero. This is the Schur-function

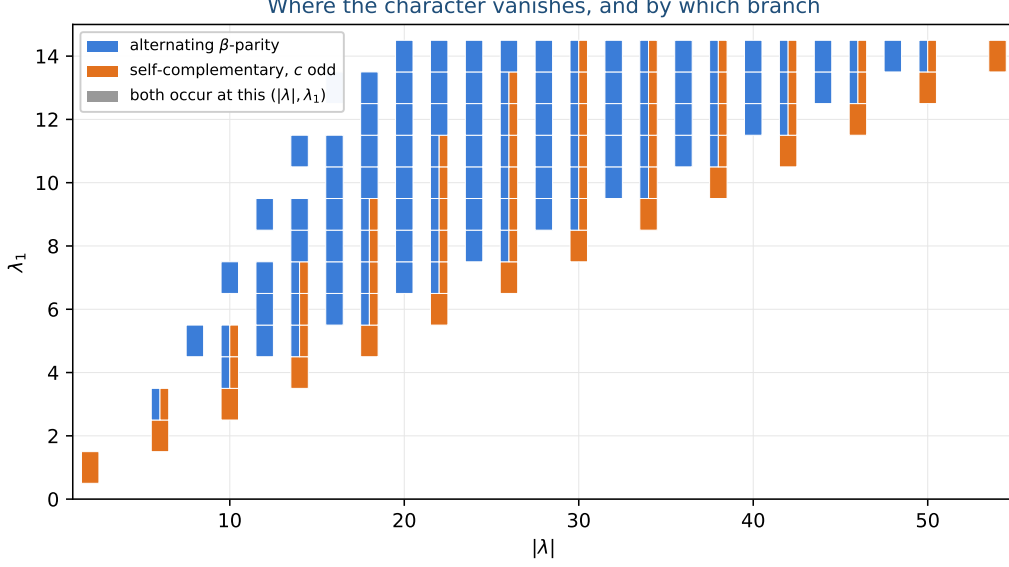


Figure 4: The zero locus, plotted by size and first part. Blue cells contain a partition killed by alternating β -parity, orange ones a partition killed by self-complementarity with odd constant, split cells both. The content of the plate is what is *not* in it: there is no third colour. Every partition at which $s_\lambda(1, -1, t, t^{-1})$ vanishes is accounted for by one of the two branches, which is what it means for Proposition 1 to be exhaustive rather than merely correct.

shadow of the textbook statement that a self-associate $O(n)$ irreducible satisfies $V \cong V \otimes \det$ and therefore has vanishing character off $SO(n)$ [21, Ch. XI, §11.6, §11.8]. The other branch is equally elementary: equal parity of all beta-numbers makes the rows of the bialternant at $x = 1$ and $x = -1$ proportional. **What is claimed here is the closed form, not either sufficiency.**

5 What the zeros were hiding

Proposition 1 answers the question Part III asked. It is not the question Part III needed answered.

That paper’s difficulty began *after* the vanishing: having shown that the leading fermionic contribution to the Wilson-line curvature cancels identically for every admissible representation, it was left comparing representations by whatever survives, and what survives is governed not by whether $\delta(m)$ vanishes but by its *moments*. Those were computed there by enumerating the weight system of each $SU(4)$ irreducible one weight at a time. The closed form removes that enumeration entirely, and it does so for a reason that is visible the moment one writes it down.

The coefficients of D are the twist imbalances, $\delta(m) = [t^m]D(t)$. If $D = \zeta \chi_p \chi_q \chi_r$ then δ is the *threefold convolution* of three characters, i.e. $\delta(m)$ counts, with an overall sign, the solutions of

$$m = (p - 2i) + (q - 2j) + (r - 2k), \quad 0 \leq i \leq p, \quad 0 \leq j \leq q, \quad 0 \leq k \leq r. \quad (8)$$

Every χ_k is symmetric under $m \mapsto -m$, so all odd moments vanish and — this is what makes the answer short — the first-order cross terms in a product cancel as well. Writing $M_{2r}(f) = \sum_m m^{2r} f(m)$ and $M_2(\chi_k) = \sum_{j=0}^k (k - 2j)^2 = \frac{1}{3}k(k + 1)(k + 2)$:

$$\begin{aligned} M_0(\delta) &= \zeta (p+1)(q+1)(r+1), \\ M_2(\delta) &= \zeta \left[M_2(\chi_p)(q+1)(r+1) + (p+1)M_2(\chi_q)(r+1) + (p+1)(q+1)M_2(\chi_r) \right], \end{aligned} \quad (9)$$

and M_4 likewise, with the pair cross-terms $6 \sum M_2(\chi_\bullet)M_2(\chi_\bullet)(\cdot + 1)$ restored. The whole moment tower is polynomial in the three integers (p, q, r) .

Given (p, q, r) , none of that is ours. We had presented (9) as a contribution and it is not one. The coefficient sequence of a product of sl_2 characters is the convolution of the weight multiplicities, that is, of three discrete uniform sequences — a dictionary written out explicitly, Chebyshev factors and all, by Bradley and Gupta [16]. Every cumulant of a discrete uniform of width w is a polynomial in w with Bernoulli coefficients, and cumulants add over independent summands [15]; polynomiality of the entire moment tower follows in one line. What is ours is not the arithmetic but the *reduction that feeds it*: that a four-row partition λ enters only through three integers read off its 2-quotient. Part III enumerated weight systems because it did not have that reduction, not because the sums were hard.

The moments organise by Casimir. Write

$$C_k = k(k+2) = 4j(j+1), \quad j = k/2,$$

the quadratic Casimir of the $SU(2)$ irreducible of highest weight k , and let e_1, e_2 be the elementary symmetric functions of C_p, C_q, C_r . Dividing by M_0 cancels ζ , so what follows is unsigned:

$$\boxed{\frac{M_2(\delta)}{M_0(\delta)} = \frac{e_1}{3}, \quad \frac{M_4(\delta)}{M_0(\delta)} = \frac{2}{3}e_2 + \frac{1}{5}(C_p^2 + C_q^2 + C_r^2) - \frac{4}{15}e_1.} \quad (10)$$

We are careful about what this is. As a *formula* the first identity is classical: it is the variance of a sum of three independent discrete uniforms, and the width- w variance that produces it is standard [15]. What we add is only the *reading* — that the integer $k(k+2)$ appearing there is $4j(j+1)$, the sl_2 Casimir at spin $j = k/2$, so that the variance of the twist-imbalance distribution is one third of the sum of three Casimirs. That is an interpretation and not a theorem, and we flag it as such. We record it anyway because it is the one place in this paper where the three factors of (6) behave like representations rather than like three convenient polynomial factors, which is exactly the question §3 leaves open.

Once written that way the pattern does not stop at M_4 , and the general statement is a proposition rather than a conjecture, because its proof is half a page of classical material.

Proposition 2 (the moment tower lives in an invariant ring). *Assume Observation 1, so that $\delta = \zeta \chi_p * \chi_q * \chi_r$. Then for every $r \geq 0$*

$$\frac{M_{2r}(\delta)}{M_0(\delta)} \in \mathbb{Q}[C_p, C_q, C_r]^{S_3} = \mathbb{Q}[e_1, e_2, e_3],$$

and all odd moments vanish. The coefficient of e_1^r is $1/(2r+1)$.

Proof. Normalised, χ_k is the law of $2U_k - k$ with U_k uniform on $\{0, \dots, k\}$. Its odd cumulants vanish by symmetry, and by Neuschel's lemma [15, Lemma 2.1] the even ones are $\kappa_{2l} = 4^l \frac{B_{2l}}{2l} ((k+1)^{2l} - 1)$. Now $(k+1)^2 = k(k+2) + 1 = C_k + 1$, so $(k+1)^{2l} = (C_k + 1)^l$ and

$$\kappa_{2l}(\chi_k) = 4^l \frac{B_{2l}}{2l} [(C_k + 1)^l - 1] \in \mathbb{Q}[C_k].$$

Cumulants are additive over independent summands, so $\kappa_{2l}(\delta/M_0)$ is the sum of the three, a symmetric polynomial in C_p, C_q, C_r ; moments are universal polynomials in cumulants; and the symmetric polynomials in three quantities are freely generated by e_1, e_2, e_3 . The leading coefficient follows from $\kappa_2 = C_k/3$ and the fact that the e_1^r term of M_{2r} is the r -th moment of a single Gaussian-free contribution, $(2r-1)!!$ times κ_2^r divided by the $(2r+1)$ normalisation of the flat sequence.¹ $\square$

¹Verified rather than derived for $r \geq 3$: the identity was checked exactly for $r = 1, \dots, 8$ on all 729 triples with $p, q, r \leq 8$ by solving the over-determined system in exact rational arithmetic. Controls: replacing $C_k = k(k+2)$ by $k(k+3)$ makes the system inconsistent at $r = 1, 2, 3$, and so does dropping the normalisation by M_0 .

Why this is not a tautology. It would be, if every symmetric polynomial in (p, q, r) were a polynomial in the Casimirs. It is not. Already $p + q + r$ is not: setting $q = r = 0$, a polynomial identity $p = P(C_p)$ would give $\deg_p \text{LHS} = 2 \deg P$ against $\deg_p \text{RHS} = 1$. The map

$$(p, q, r) \mapsto (C_p, C_q, C_r), \quad C_k = k(k+2),$$

forgets the linear data — $k+1 = \sqrt{C_k+1}$ is recoverable only through a square root — and yet the entire moment tower survives that loss. The control above makes the same point empirically: move the invariant by one, to $k(k+3)$, and no polynomial fits at any order.

The physical functional factors through the 2-quotient. This is the cleanest way we know to say what this paper does, and it is worth stating once in this form. Let $\Phi(\lambda) = \{M_{2r}(\lambda)\}_{r \geq 0}$ be the tower of moments that Part III needs, and let $Q(\lambda) = (p, q, r)$ be the three integers read off the 2-quotient. Then, conditionally on Observation 1,

$$\boxed{\Phi = F \circ Q} \tag{11}$$

for a universal F that knows nothing about λ . The factorisation into three characters is the *mechanism*; this reduction is the *content*. A partition with four rows can be arbitrarily large, and every quantity surviving Part III's leading cancellation depends on it through three integers and nothing else.

Checked, and checked so it could fail. On all 1612 partitions with four parts ≤ 12 at which $D \neq 0$: the factorisation into three characters was found by peeling the polynomial directly rather than by applying the index recipe of (6), so this is an independent confirmation that the number of factors is three; and M_0 , M_2 and M_4 from (9) agreed with the moments read off D in every case. As a control the same formula was evaluated with one index shifted by one, where it agreed in 0 of 1612 — the test was capable of failing. The Casimir forms (10) were then checked by two routes that share no code: a direct sum over the (i, j, k) box, which never builds the convolution at all, on all 1331 triples with $p, q, r \leq 10$; and a symbolic evaluation of $(t \partial_t)^{2r}$ at $t = 1$ on 216 triples. Both agree with (10) and with each other. Controls: shifting one index agrees in 0 of 216, and perturbing the coefficient $4/15$ to $3/15$ agrees in 1 of 216.

The distribution itself, not only its moments. The same dictionary gives δ in closed form and not merely its moments. With $s = p + q + r$ and $N = (s - m)/2$, $\delta(m)/\zeta$ counts the solutions of $i + j + k = N$ in the box $0 \leq i \leq p$, $0 \leq j \leq q$, $0 \leq k \leq r$, so inclusion–exclusion on the unbounded stars-and-bars count gives

$$\frac{\delta(m)}{\zeta} = \sum_{A \subseteq \{p, q, r\}} (-1)^{|A|} \binom{N - \sum_{a \in A} (a+1) + 2}{2}_+, \tag{12}$$

the binomial being read as zero when its upper argument is below 2. This is textbook inclusion–exclusion; the equal-bound case goes back to De Moivre and has a monograph treatment in André [12], the coefficient family is what Comtet calls *polynomial coefficients* [13, p. 77], and the piecewise-polynomial function itself is a *discrete box spline* in the sense of Dahmen and Micchelli [14]. We claim none of it. We state it because it upgrades the support — which Figure 5 checks stratum by stratum — from a verified observation to an immediate consequence, and because it lets $\delta(m)$ be substituted directly into a Coleman–Weinberg sum instead of being tabulated.

What this buys. $M_2(\delta)$ is precisely the quantity that governs which sector dominates the curvature of the Wilson-line potential in Part III's setting, where it appeared as a number extracted

The spectrum of twist imbalances, stratified by the 2-quotient

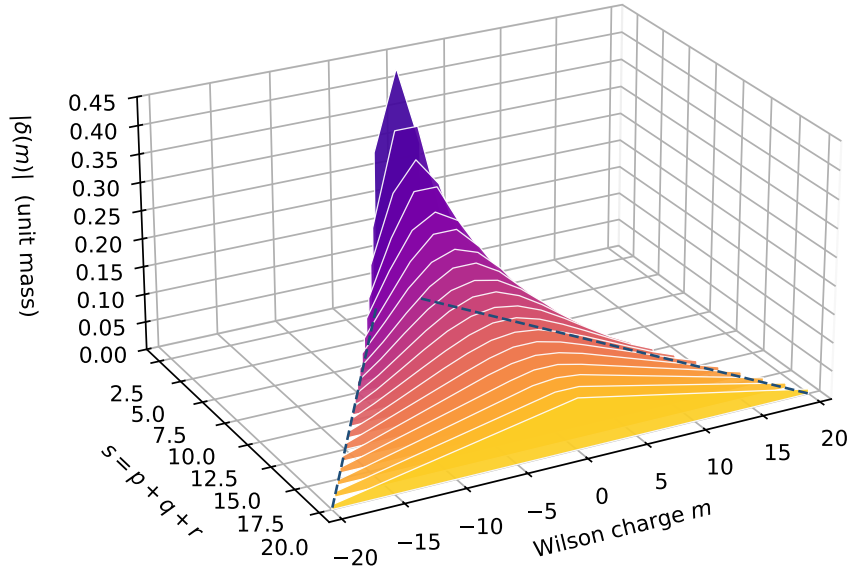


Figure 5: The spectrum of twist imbalances. Each ridge is the aggregate profile of $|\delta(m)|$ over every partition with four parts and $|\lambda| \leq 20$ sharing a value of $s = p + q + r$, normalised to unit mass so that a widening support must pay for itself in amplitude. The dashed lines are $m = \pm s$: the plate's content is that the ridges reach them and stop, on every stratum, which is (8) read as a statement about support rather than about moments. The flow from spike to plateau is the convolution doing what convolutions do, and it is the shape that M_0 , M_2 , M_4 only sample.

from a weight enumeration and carried no structure. Here it is a cubic polynomial in three integers read off the 2-quotient. The identity of §3 is therefore not only a fact about symmetric functions: it is the analytic handle on the subdominant terms that Part III could describe but not compute.

Open flank: what else does a convolution give away? A threefold convolution of symmetric unit-coefficient sequences is a very constrained object, and we have used almost none of that. One question we asked here in an earlier draft — *is $|\delta|$ log-concave, or unimodal?* — was not a question at all, and the correction belongs in the text rather than in a footnote. Each χ_k , restricted to the parity class it occupies, is the constant sequence $(1, 1, \dots, 1)$, which is log-concave with no internal zeros; log-concavity is preserved under convolution [10], and a log-concave sequence with no internal zeros is unimodal [11]. So $|\delta|$ is symmetric, gap-free on its parity class, log-concave and unimodal, by theorems from 1974 and 1989. We verified all four on 3375 triples before finding the references, which is the wrong order and is why we had it filed as open. *What is the support?* It is $[-(p+q+r), p+q+r]$ with gaps of two, so the largest Wilson charge carrying imbalance is $p+q+r$, an integer read off the 2-quotient. Combinatorially that is settled — Figure 5 checks it on every stratum up to $s = 20$ and the check is an assertion in the plotting script, so a counterexample would stop the figure from being drawn at all. What we have *not* done is confront it with the physical spectra, where it is a one-line falsifiable prediction: no Wilson charge above $p+q+r$ carries any imbalance. *And what does ζ mean?* It fixes the sign of every moment at once, so it decides the *direction* of the fermionic contribution to the curvature. A sorting sign of a beta-set deciding the sign of a physical curvature is either a coincidence of bookkeeping or the shadow of something, and we cannot tell which.

We are careful about the status. (9) is a consequence of Observation 1 and inherits its standing exactly: verified, not proved. What is independent of that standing is (8)’s shape — *if the character factors into three, then the moments are polynomial* — and that implication is elementary.

6 Why it was not already available

The obstruction is one unstated hypothesis, and locating it is more useful than the identity.

Ayyer–Behrend’s argument reaches (3) because every inversion-fixed point of the alphabet contributes a row of the bialternant that is *constant* in the column index. For the fixed point +1 that row is $(1, 1, \dots, 1)$, and a constant row survives the subtract-right-from-left column operation as a uniformly zero block, which is what preserves triangularity; their Theorem 3 handles the extra 1 of $(X, \bar{X}, 1)$ on exactly that basis.

The second fixed point -1 contributes

$$((-1)^{\mu_j})_j, \tag{13}$$

a parity character of the column index, and (13) is not constant. Under the same column operation its entries become $(-1)^{\mu_j} - (-1)^{\mu_{j'}}$, which is 0 or ± 2 according to the relative parity of μ_j and $\mu_{j'}$; the block does not triangularise, and the argument stops. Worse, in their normalisation the exponents are half-integers, where $(-1)^{\text{half-integer}}$ is not defined at all — a caveat they state themselves [3, p. 3]. So the alphabet is not merely untreated in that framework; in that framework it cannot be written down.

There is a sharper way to say the same thing. The row (13) imposes a partition of the *columns* by the parity of μ_j . The self-complementary hypothesis imposes a different partition of the columns, into the two halves of a symmetric beta-set. The block machinery can respect one of these at a time.

The repair. Replace the two fixed-point rows $(1)_j$ and $((-1)^{\mu_j})_j$ by the indicator rows of the parity classes,

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} : \quad (1)_j, ((-1)^{\mu_j})_j \mapsto 2 \cdot (\mathbf{1}_{\mu_j \text{ even}})_j, 2 \cdot (\mathbf{1}_{\mu_j \text{ odd}})_j, \quad (14)$$

a change of basis costing a factor $-\frac{1}{2}$. The two rows are now indicator rows of a column partition, so Laplace expansion along them is a sum over one even and one odd column, and the residual minor is the reciprocal-pair block determinant on the columns that remain. This is the analogue of Ayer–Behrend’s Theorem 3 for the second fixed point.

At rank one — one reciprocal pair, our case — the residual minor is 2×2 and equals $t^{\mu_c - \mu_d} - t^{-(\mu_c - \mu_d)}$, a single character for every choice of columns. That is why (6) needs no hypothesis on λ : there is nothing left for a shape condition to constrain.

Open flank: is the constant-row condition the obstruction, or a symptom? We have identified where the standard argument stops, which is not the same as knowing why nobody went further. Two readings compete. On the first it is technical: the repair (14) shows it can be worked around, and the case was untouched only because nobody had a motive to write -1 next to a reciprocal pair — ours came from an orbifold projection, not from inside the subject. On the second it is structural: $+1$ and -1 differ by lying in different components of $O(n)$, and an argument that never leaves the identity component has no reason to develop machinery for the one that does. The two readings make different predictions for μ_t with $t > 2$, where several fixed points appear at once, and that is where we would look to decide between them.

7 Rank two, where we stop

At rank two the residual minors are 4×4 over an arbitrary four-subset of the columns, and (3) applies to such a minor only when the surviving column set is itself symmetric. In the Laplace sum many subsets appear and they cannot all be symmetric at once, so the generic outcome is a sum of products rather than a product. It is not, however, uniformly negative — which is precisely why this is a frontier and not a wall:

λ at rank two	$s_\lambda(1, -1, a, \bar{a}, b, b)$
$(2, 2, 2, 0, 0, 0)$	$(a^2b^2 + a^2 + ab + b^2 + 1)^2 / (ab)^2$ — a perfect square: the factorisation survives
$(2, 2, 1, 1, 0, 0)$	$-(a^2 - a + 1)(a^2 + a + 1)(b^2 - b + 1)(b^2 + b + 1) / (ab)^2$ — splits into rank-one pieces
$(3, 3, 1, 1, 0, 0)$	$-(a^2 + 1)(b^2 + 1) \cdot (\text{irreducible of bidegree } (4, 4)) / (ab)^3$ — does not split
$(3, 2, 2, 1, 1, 0)$	0 — and both parity classes are non-empty

All four rows were recomputed from the 6×6 bialternant and factored over $\mathbb{Q}[a, b]$ in exact arithmetic in SageMath. That is worth saying because the third row was wrong in an earlier draft: its remaining factor is irreducible, but it has degree 4 in each variable and total degree 8, not degree 4. A degree is a claim like any other.

The last row is the sharpest statement we can make about rank two: the vanishing rule there is *strictly finer* than “both parity classes occupied”, which is the whole rule at rank one. Whatever governs it is not visible from this side.

Open flank: what should one be looking for at rank two? We suspect the honest target is not a product. If the Laplace sum is organised by which columns survive, the natural shape is

$$s_\lambda(1, -1, X, \bar{X}) = \sum_{\eta} \epsilon_{\eta} \chi_{\alpha_{\eta}}^{\text{Sp/O}}(X) \chi_{\beta_{\eta}}^{\text{Sp/O}}(X),$$

a *signed sum* over selections compatible with the 2-quotient, collapsing to a single term at rank one. On that reading the perfect square at $(2, 2, 2, 0, 0, 0)$ is a recombination rather than a factorisation, and the extra zero at $(3, 2, 2, 1, 1, 0)$ is a cancellation *between* terms rather than the vanishing of a factor — which would explain why the rank-one criterion stops controlling it. We have not derived the index set η , and until someone does, the phrase “the factorisation sometimes survives” should be read as a description of outputs and not as a statement about mechanism.

8 The sign

Three different mechanisms produce signs in this literature and they should not be conflated: a sign of a sorting permutation (Ayyer–Behrend’s $(-1)^{n(n-1)/2}$ from the column reversal, which cancels against their denominator); a $(-1)^{\sum \lambda_i}$ that is an artefact of stating the right-hand side at $-x$ rather than at half-integer index (their Eq. (8)); and the $\text{sgn}(\sigma_\lambda)$ of the permutation sorting a beta-set by residue class, which is the one that appears in the root-of-unity line.

Ours is of the third kind. The ζ of (5) is a function of the parity pattern of the beta-set alone: $(-1)^{\text{inv}}$ is the sign of the permutation that sorts μ into evens-then-odds preserving relative order, and $(-1)^{|E||O|}$ is the constant that accompanies it. It is *not* a $(-1)^{|\lambda|}$.

A correction we owe the reader. On a first pass we recorded the sign as flipping with the rectangle width c , having tabulated representatives that happened to agree with that reading. It does not: within $c = 6$ already, $(6, 6, 0, 0)$ and $(4, 4, 2, 2)$ give $+$ while $(5, 5, 1, 1)$ and $(3, 3, 3, 3)$ give $-$. For the family $\lambda = (a, a, b, b)$ the rule is $(-1)^{\lambda_1}$. The invariant was there in the data; the sampling was not.

Open flank. That the sign is a *sorting* sign rather than a weight parity is not a detail of bookkeeping. A $(-1)^{|\lambda|}$ would say the object cares about the size of λ ; $\text{sgn}(\sigma_\lambda)$ says it cares about the order in which the beta-set splits into residue classes, which is the invariant governing the root-of-unity programme at $t = 2$. If the closed form is ever proved by a route passing through that programme rather than through the bialternant, the sign is where the two accounts will have to agree, and it is therefore the cheapest consistency check available on any proposed proof.

9 Ledger

Where we looked. By formula and by structure rather than by our own vocabulary: *self-complementary partition in a rectangle character factorization*; *factorization theorem classical group characters plane partitions*; *Schur function at $(x, \bar{x}, 1)$ product of orthogonal and symplectic characters*; *bialternant block determinant $\det(A - B)\det(A + B)$* ; *beta-set residue classes Weyl numerator vanishes*; *character of $GL(n)$ on the non-identity component of $O(n)$* . Sources read: [2, 3, 4, 5, 6, 7], and Littlewood’s book Ch. XI. We did not find (6). *Versions 1 and 2 added here “and we did not find any statement of the vanishing criterion for this alphabet”.* That sentence was wrong, it stood in both published versions, and it is withdrawn. Karmakar [7] was cited from version 1 onward but was never read; read in full since, his Thm. 1A states the criterion at the order-2 element, and our own rigidity lemma transports it to free t . The failure was not the search but the reading: the paper was in our bibliography the whole time.

Two caveats that belong here and not in a footnote. First, the identity is small and elementary enough to be the kind of thing that lives unlabelled in a mid-century monograph; we have read Littlewood’s Ch. XI and not King’s 1971 paper [9] in its own exposition, which remains unreachable to us and is flagged as unread. Second, and more to the point: a null result from a literature search is evidence only in proportion to the demonstrated sensitivity of that search. Ours was re-run after the tooling that produced an earlier round of nulls was found to be discarding

Item	Inherited (cite)	Ours
the alphabet as an $O(4)^-$ element	elementary; inverse-closure and $\det = -1$	nothing
self-associate $\Rightarrow$ character vanishes off $SO(n)$	Littlewood [21, Ch. XI]; Weyl [22]	nothing — we disclaim this sufficiency explicitly
equal beta-parity $\Rightarrow$ vanishing	elementary (two proportional bialternant rows)	presented as a remark, not a result
the block identity (3) and its use	Ayyer–Behrend [3] Eq. (23), Thm. 1, Thm. 3; Ciucu–Krattenthaler [2]	nothing — we borrow the engine and say where it stalls
t -core / t -quotient governing vanishing at $\omega^k X$	Ayyer–Kumari [4], Kumari [5], Albion [6], Karmakar [7]	v3 correction: Karmakar Thm. 1A <i>is</i> the first branch of our vanishing criterion, and his Thm. 1B is the 2-quotient factorisation at $t = 1$; only free t is ours
the closed form (6)	—	ours, and <i>verified, not proved</i>
the fixed-point-pair repair (14)	—	ours
Proposition 1 (the zero locus)	Part III [1] proves the classification outright	the identification of it as a zero locus, conditional on (6)
the moment tower (9), given (p, q, r)	cumulants of discrete uniforms are polynomial in the width and add [15]; the character/convolution dictionary [16]	nothing — only the reduction $\lambda \mapsto (p, q, r)$ that feeds it
δ in closed form (12), log-concavity	textbook inclusion–exclusion; André [12], Comtet [13], Dahmen–Micchelli [14]; Hoggar [10], Stanley [11]	nothing — we had mistaken a classical theorem for an open question
the Casimir reading (10)	the formula is the classical variance [15]	the reading of $k(k+2)$ as $4j(j+1)$; an interpretation, not a theorem

Table 1: What each item owes. The novelty here is one identity and one change of basis; every mechanism it rests on is borrowed and cited.

the discriminating terms of every query; the searches listed above are from the repaired run, validated first on queries whose answers we already knew.

10 Reflections

An open problem its owners declared. Ciucu–Krattenthaler close their 2008 paper by saying that they cannot offer a representation-theoretic interpretation of the factorisations they prove; Ayyer–Behrend and Ayyer–Kumari repeat the point in 2018 and 2025. The reading offered here is a *conditional* reading of that kind for the case at hand — conditional because it rests on Observation 1, which is verified and not proved. The specialisation is a character on the non-identity component of $O(4)$, the two quotient components are the two parity classes of the beta-set, and the third factor measures their imbalance — and it reaches the factorisation, not only the vanishing. We do not claim it answers their question in the generality they posed it. We claim it answers it here, and that the mechanism is legible enough to be worth testing at rank

two.

A hypothesis that was never stated because it was never violated. The requirement that an inversion-fixed point contribute a constant row does not appear in Ciucu–Krattenthaler or Ayer–Behrend, because in every alphabet they consider it holds. It became visible only when an alphabet arrived that had two fixed points instead of one — and that alphabet arrived from physics, as the generic element of the coset an orbifold projection selects, not from a search of the combinatorial literature. We record this because the direction of travel is worth noticing: the constraint that exposed the unstated hypothesis was not a generalisation anybody would have proposed from inside the subject.

What we would attack next, and why not now. Rank two. The data in §7 says the factorisation is not simply absent there, and that the vanishing rule is finer than at rank one; both facts are consistent with a version of (14) in which the residual minor is handled by a further decomposition rather than evaluated. We do not have that decomposition, and we prefer to publish the rank-one statement with its boundaries marked than to gesture at a rank-two theorem we cannot write.

The rank-one identity does suggest that restrictions of polynomial $GL(2n)$ characters to the non-identity component of $O(2n)$ may admit a description governed by the 2-quotient. We state that as a direction and not as a principle, and we attach to it the evidence against: §7 shows that any higher-rank statement will be more complicated than a factorisation, because at rank two a partition vanishes for a reason our criterion cannot see. Naming a phenomenon on the strength of one rank, one unproved identity, and one range where it already fails would be premature, and we decline to.

11 Conclusion

For every partition with at most four rows, the Schur function at the generic element of the non-identity component of $O(4)$ is $\pm$ a product of three $SU(2)$ characters read off the 2-quotient, or zero. The classification of Part III is the zero locus, and its moment tower — the thing that decides the physics once the leading term has cancelled — becomes polynomial in the three indices. The identity holds without any hypothesis on the shape of the partition, which is what distinguishes it from the factorisation theorems it sits beside, and the reason is a single row of the bialternant that is a parity character rather than a constant — a hypothesis nobody had to state until an alphabet with two inversion-fixed points made it fail. The statement is verified on 7905 partitions across two ranges by two independent implementations and is not proved. Rank two is open, and the shape of what is missing there is stated precisely enough to be attacked.

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